

BABY CRY AUTO DETECTOR WITH MUSIC PLAYER

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DECLARATION

I declare that this project proposal is my own work and has never been presented in part or wholly to any institution for the award of any academic qualification

NAME:

SIGNATURE:

DATE:

Oro Vinton Daniel.

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DECLARATION BY SUPERVISOR

This project proposal has been submitted with my approval at kabete national polytechnic.

NAME:

SIGN:

DATE:

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ABBREVIATIONS OR SYMBOLS

LED- Light emitter diode

PIC- peripheral interface controller

IC- integrated circuit

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ABSTRACT

Automatic voice detection of baby cry will be playing an outstanding role in different applications for smart monitoring of smart baby condition. In this proposed model, a baby's cry will be detected and a music player will be played after detection in order to create a soothing environment for the baby. This system will employ a machine learning approach to recognize newborn cry sounds in a variety of residential settings under difficult situations. The automatic detection of a baby cry will be used in a variety of situations involving various types of sounds in the environment. The proposed system will also be provided with alert-based notifications to parents. Also, in commercial used products such as remote monitoring of baby, baby facial recognition as well as in medical applications. Also, evaluate K-Nearest Neighbours (KNN) algorithm for baby cry detection and perform priority queue functions for playing music. In this system, the input will consist of the cry sound with k nearest training samples in the database. The output will be depended on analysis of the cry-detection performance with KNN which is used for classification.

CHAPTER 1

1.0 Introduction/background

Parents in the present world are busy in their professional life, so they do not get sufficient time to take care of their babies. It is inconvenient for parents to constantly watch over their newborn baby while doing their work or chores. So we have designed a simple system which helps parents in taking care of baby.

This system proposes a simple voice detection system which can be applied practically for designing a device with capability to detect a baby's cry and automatically turning on a Baby sleep music.

The system is based on PIC microcontroller. Whenever the baby cries it is detected by the controller with the help of Mic and in order to that turns on the music and sleep mechanism which creates soothing sound as well as dim lights which makes baby sleep again gently. In this project a program is implemented to detect an infant's crying. It can detect baby's cry while ignoring other sounds like clap, sneeze, fan, sudden sounds, environment sounds etc

1.1 Problem statement

It is inconvenient for parents to constantly watch over their newborn baby while doing their work or chores because in the present world parents are too busy with their professional lives. So we have designed a simple system which will detect baby's cry and start playing a soothing music to help the baby sleep which will help parents in taking care of baby.

1.2 Justification

This research on baby cry auto detector with sleep music player is necessary since it helps by detecting baby crying and plays soothing music to help the baby sleep, also help the parents in monitoring their babies or whenever they are occupied

The system is cheap and will help parents in baby sitting

1.3 objectives

1.3.1 main objectives

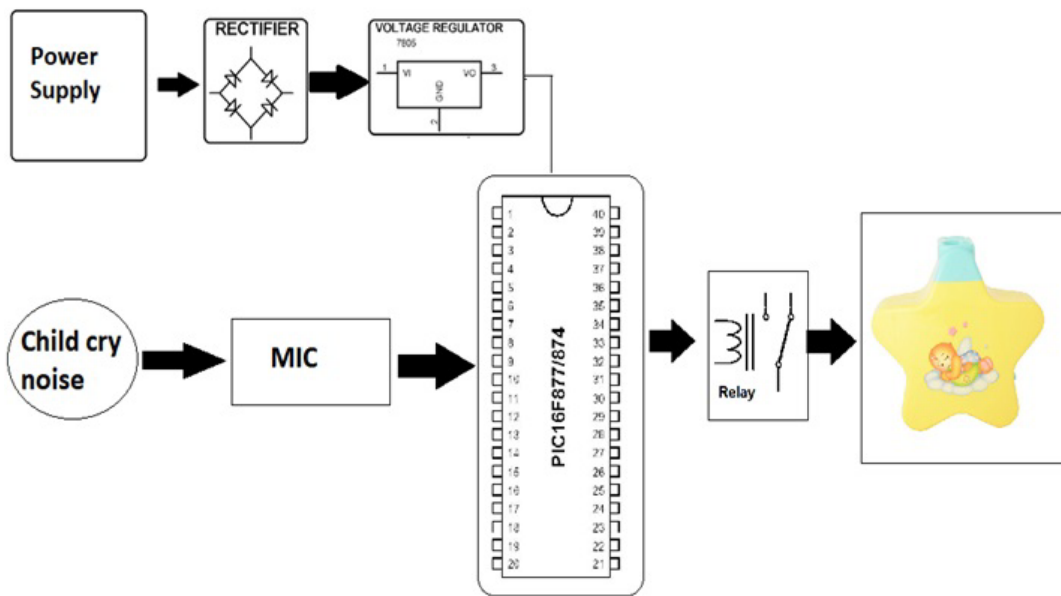
To design and implement PIC micro controllers

1.3.2 specific objectives

To test the operation of the PIC micro controllers

To automatically detect cries

1.4 Block diagram



1.5 components and materials

PIC micro controller.

mic sensor.

Relay.

Crystal Oscillator.

Resistors.

Capacitors.

Transistors.

Cable and connectors

Diodes

Transformer / adapters

Push button

Switch

IC

IC sockets

LED

PCB and breadboard

